

Masdar - Qatar University Academic Resource Platform

A centralized platform for Qatar University students to access, share, and request educational resources including lecture notes, past exams, papers, and course materials.

Why I Built This

As a student at Qatar University, I noticed a recurring problem that affected me and my peers. We were all struggling to find academic materials - lecture notes, past exams, study guides - because everything was scattered across different platforms, WhatsApp groups, and personal collections. There was no central place where we could reliably access or share these resources.

I saw students spending hours searching for materials that someone else already had. I watched as valuable resources got lost because there was no organized system to preserve them. Course materials that could help future students were trapped in individual devices or forgotten group chats.

That's when I decided to build **Masdar** - a platform that would solve these problems for our university community.

What Masdar Does

I designed Masdar to be a comprehensive solution where:

- **Everything is in one place:** All course materials are centralized and easy to access, so you don't have to hunt through multiple channels to find what you need.
- **Students contribute together:** The platform is community-driven. Anyone can submit resources for existing courses or request new courses to be added, making it a collaborative space.
- **Navigation is simple:** I built an intuitive interface that lets you browse courses by department and quickly find the materials you're looking for.
- **It's secure and scalable:** I implemented JWT authentication for administrative functions and used cloud-based storage to ensure the platform can grow with our needs.
- **Downloads are fast:** By leveraging Cloudflare R2 for storage, I ensured that accessing and downloading content is quick and reliable.

Technical Architecture

Tech Stack

- **Backend:** Node.js with Express.js
- **Database:** MongoDB for storing course and resource metadata
- **Storage:** AWS S3 / Cloudflare R2 for file storage
- **Authentication:** JWT-based authentication for admin panel
- **Deployment:** Vercel for seamless hosting
- **Frontend:** Vanilla JavaScript with modern ES6+ features

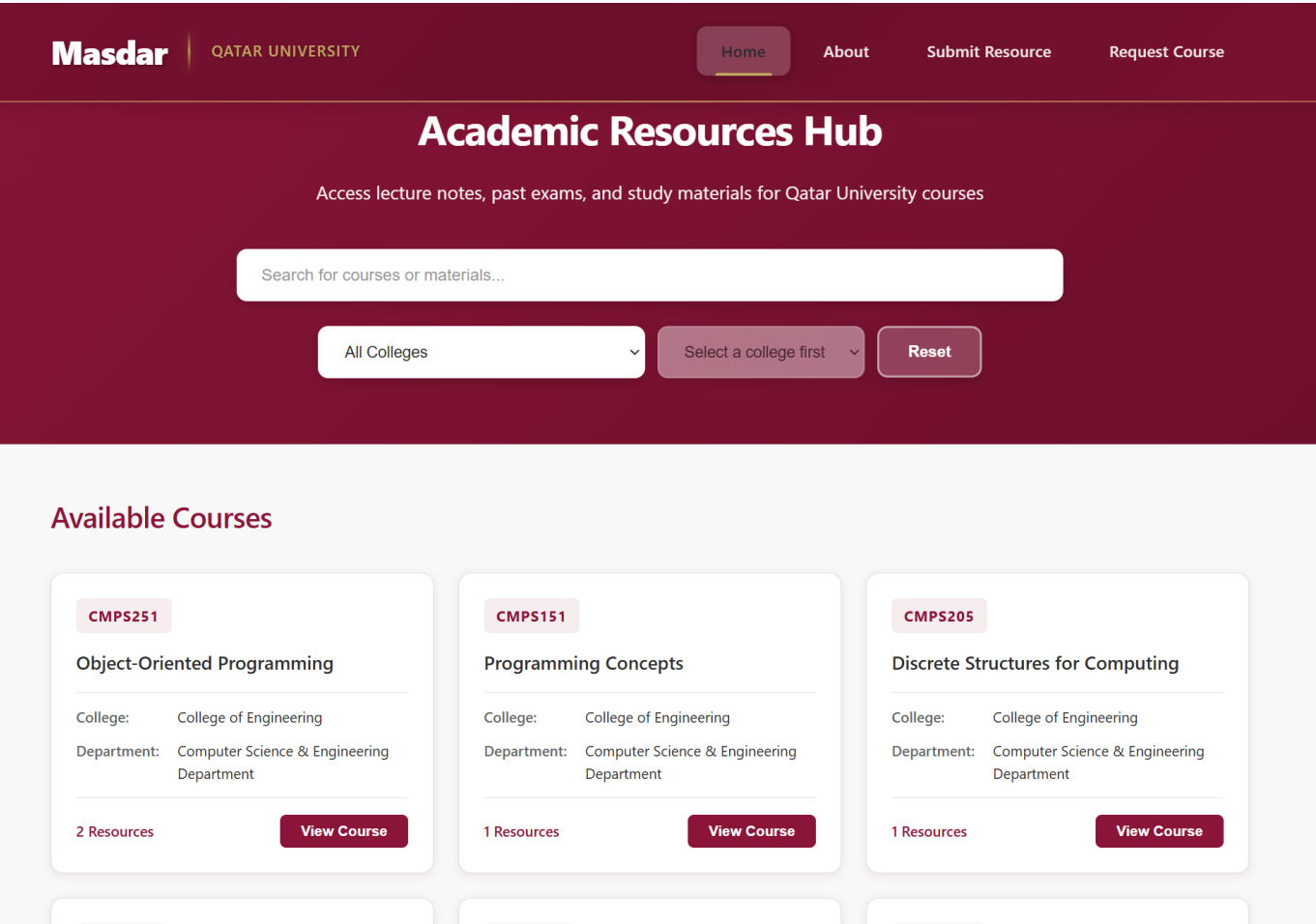
Key Features

I built the following features to make the platform as useful as possible:

- Browse courses organized by department and category
- Download lecture notes, past exams, and papers with one click
- Submit your own resources to help other students
- Request new courses that aren't yet available
- Admin panel with secure authentication for content moderation
- Fast content delivery through CDN integration

Application Screenshots

Home Page



Browse all available courses and resources organized by department

Course Details

Masdar

QATAR UNIVERSITY

Home

About

Submit Resource

Request Course

Object-Oriented Programming

2 Resources

+ Add Resource for this Course

COLLEGE

College of Engineering

DEPARTMENT

Computer Science & Engineering Department

Resources

All Types

All Years

OOP Exams

EXAM

Download

First, Second, and Final Exams for OOP

Year: 2025

10 downloads

Exams

Questions

Lecture Notes for OOP

LECTURE NOTES

Download

OOP Lecture Notes for Studying

View detailed information about courses and download available materials

Submit Resources

Masdar

QATAR UNIVERSITY

Home

About

Submit Resource

Request Course

Submit a Resource

Help fellow students by sharing educational materials. Your submission will be reviewed before being published.

College *

Select College

Don't see your College? [Request it here](#)

Department *

Select College First

Don't see your Department? [Request it here](#)

Course Code *

Type to search courses...

Don't see your course? [Request it here](#)

Resource Title *

e.g., Midterm Exam - Fall 2024

Resource Type *

Select Type

Students can contribute by submitting resources for existing courses

Request New Courses

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HomeAboutSubmit ResourceRequest Course

Request a New Course

Don't see your course listed? Request it here and an admin will add it to the database.

College Name *

Select College

Select the college

Department Name *

Select College First

Select the department

Course Code *

e.g., CMPS251

Format: 4 letters + 3 numbers (e.g., CMPS251)

Course Name *

e.g., Data Structures and Algorithms

Enter the full course name

Submit Request

Cancel

Request courses that aren't yet available on the platform

About Page

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HomeAboutSubmit ResourceRequest Course

About Masdar

Masdar is a student-created platform designed to help Qatar University students easily access and share educational resources such as past exams, lecture notes, and study materials.

Our Mission

We believe that education is enhanced when students collaborate and share knowledge. Masdar provides a centralized hub where students can contribute and access quality academic resources, making learning more efficient and collaborative.

What We Offer

- Past exam papers and solutions
- Lecture notes and study guides
- Course assignments and projects
- Easy-to-navigate course catalog

Learn more about the platform's mission and features

Getting Started

Prerequisites

- Node.js (v14 or higher)
- MongoDB instance
- AWS S3 or Cloudflare R2 account
- Environment variables configured

Installation

1. Clone the repository

```
git clone https://github.com/Ahmed-Elsheshtawy/qu-educational-content.git
cd qu-educational-content/source
```

2. Install dependencies

```
npm install
```

3. Configure environment variables

Create a `.env` file in the source directory:

```
MONGODB_URI=your_mongodb_connection_string
JWT_SECRET=your_jwt_secret_key
AWS_ACCESS_KEY_ID=your_aws_access_key
AWS_SECRET_ACCESS_KEY=your_aws_secret_key
AWS_REGION=your_aws_region
S3_BUCKET_NAME=your_bucket_name
```

4. Start the server

```
npm start
```

5. Access the application Open your browser and navigate to <http://localhost:3000>

Project Structure

```
source/
├── server.js           # Main Express server
├── package.json        # Project dependencies
├── vercel.json         # Vercel deployment config
├── public/
│   ├── frontend/      # Client-side application
│   │   ├── index.html
│   │   ├── scripts/   # JavaScript modules
│   │   ├── styles/    # CSS stylesheets
│   │   └── views/      # HTML pages
│   ├── middleware/    # Authentication & JWT middleware
│   ├── routes/         # API route handlers
│   └── services/       # Business logic & external services
```

Security

I took security seriously while building this platform:

- Implemented JWT-based authentication for all administrative functions
- Used secure cookie handling for session management
- Set up environment-based configuration to protect sensitive credentials
- Protected admin routes to ensure only authorized users can moderate content

API Endpoints

Endpoint	Method	Description
/api/courses	GET	Retrieve all courses
/api/resources	GET	Retrieve all resources

Endpoint	Method	Description
/api/auth/login	POST	Admin authentication

`About the Author

My name is **Ahmed Elsheshtawy**, and I'm a student at Qatar University. I built this platform because I experienced firsthand the frustration of not having easy access to academic resources. I wanted to create something that would make student life easier and foster a culture of collaboration and knowledge sharing.

If you have questions, suggestions, or just want to connect, feel free to reach out:

- GitHub: [@Ahmed-Elsheshtawy](#)
- Project Repository: [QU Academic Content Platform](#)

License

This project is licensed under the ISC License.

Contributing

I welcome contributions from anyone who wants to improve this platform! Whether you're a student at QU or just someone who wants to help, here's how you can contribute:

1. Fork the repository
2. Create a feature branch (`git checkout -b feature/YourFeature`)
3. Commit your changes (`git commit -m 'Add your feature'`)
4. Push to the branch (`git push origin feature/YourFeature`)
5. Open a Pull Request

I'll review your contribution and work with you to get it merged.

Acknowledgments

I want to thank my fellow Qatar University students for inspiring this project and providing valuable feedback. I also want to acknowledge the open-source community for creating the incredible tools and libraries that made this platform possible.

Built with dedication

Made with ❤️ for Qatar University Students